

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Write short note on any four of the following

- a) Calibration (CO1)
- b) Dead zone (CO1)
- c) Drift (CO1)
- d) Capsule gauge (CO2)
- e) Sight glass method (CO3)
- f) Feed forward control system. (CO4)

Q.24 What is transfer function of a control system? Discuss feed forward and feed backward control system in detail. (CO4)

Q.25 Explain the principle, working and construction of capacitive pressure transducer. Mention its advantages and dis-advantages along with neat diagram. (CO2)

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Roll No.

3rd Sem / Chemical

Subject : Process Instrumentation and Control

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 The operation of U tube manometer is based on (CO2)

- a) Variable flow area
- b) Pressure at a stagnation point
- c) Pressure difference across the connecting ends
- d) Rotation of a turbine

Q.2 Which of the following is an undesirable dynamic characteristics of an instrument? (CO1)

- a) Reproducibility
- b) Dead zone
- c) Time lag
- d) Static error

Q.3 Which of the following is used for measurement of vacuum? (CO2)

- a) U-tube manometer
- b) Dead weight piston gauge

- c) Capacitive pressure transducer
- d) Capsule gauge

Q.4 Which of the following is system input? (CO4)

- a) Step input b) Ramp Input
- c) Pulse Input d) All of these

Q.5 Which of the following is NOT measured using orsat analyzer. (CO3)

- a) Oxygen b) Carbon dioxide
- c) Nitrogen d) Carbon monoxide

Q.6 Flapper nozzle is a/an _____ controller. (CO4)

- a) Pneumatic b) Hydraulic
- c) Electronic d) None of these

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Write any one expansion type thermometer. (CO2)

Q.8 Define precision. (CO1)

Q.9 1 atmospere = _____ pascal. (CO2)

Q.10 Name any two devices which are based on direct method of level measurement. (CO3)

Q.11 Define sinusoidal Input. (CO4)

Q.12 Name any two scales used for temperature measurement. (CO2)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain speed of response and fidelity characteristics of measuring instruments. (CO1)

Q.14 Explain bimetallic thermometer with the help of neat diagram. (CO2)

Q.15 Explain the principle working and advantages of U-tube manometer with the help of neat diagram. (CO2)

Q.16 What are indirect methods of level measurement? Discuss diaphragm box and air-purge method. (CO3)

Q.17 Discuss strip chart recorder in detail. (CO3)

Q.18 Explain the various components of a process control system. (CO4)

Q.19 Differentiate between open loop control system and closed loop control system. (CO4)

Q.20 Discuss the need and aim of measurement. (CO4)

Q.21 Explain principle and construction of optical pyrometer with the help of diagram. (CO2)

Q.22 Differentiate between Gauge pressure, absolute pressure and vacuum. (CO2)